DP Computer Science: A 2.1.3 Network Devices

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Identify and describe the functions of key network devices: **modems, routers, switches, WAPs**
- Explain how these devices work together to enable **network communication**
- Map network devices to the layers of the **TCP/IP model**

Key Vocabulary

Term	Definition
Modem	A device that connects a network to the internet by modulating/demodulating signals
Router	A device that directs data between networks and routes traffic
Switch	A device that connects devices within a LAN and directs data using MAC addresses
WAP (Wireless Access Point)	A device that provides wireless connectivity (Wi-Fi) to devices
NIC (Network Interface Card)	Hardware that allows a device to connect to a network (wired or wireless)
Gateway	A device that connects two different networks, translating protocols if needed

Term	Definition
Firewall	A security device that monitors and controls incoming/outgoing network traffic
TCP/IP Model	A four-layer model describing how data is transmitted over networks

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IB ATL Skills

Skill	Description
Communication	Communicating information and ideas effectively
Collaboration	Working collaboratively with peers
Thinking	Applying thinking skills critically
Information Literacy	Finding and interpreting information

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2. Network Devices Overview

The Core Devices

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┌─────────────────────────────────────────────────────────────┐
│                     NETWORK DEVICES                         │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│    ┌────────────┐    ┌────────────┐    ┌────────────────┐ │
│    │   Modem    │    │   Router   │    │     Switch     │ │
│    │  Connects  │    │  Directs   │    │  Connects LAN  │ │
│    │  to ISP    │    │  Traffic   │    │  devices       │ │
│    └────────────┘    └────────────┘    └────────────────┘ │
│                                                             │
```

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│    ┌──────────────┐   ┌──────────────┐   ┌──────────────────┐   │
│    │     WAP      │   │     NIC      │   │    Firewall      │   │
│    │   Provides   │   │   Connects   │   │    Security      │   │
│    │   Wi-Fi      │   │   Device     │   │    Device        │   │
│    └──────────────┘   └──────────────┘   └──────────────────┘   │
│                                                                   │
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```

Device Functions at a Glance

Device	Function	Key Feature
Modem	Connects to the internet via ISP (modulates/demodulates signals)	Converts digital to analogue and vice versa
Router	Directs data between networks; routes traffic to correct destination	Uses IP addresses; connects LAN to WAN
Switch	Connects devices within a LAN; directs data to specific devices	Uses MAC addresses; operates at Data Link layer
WAP	Provides wireless connectivity (Wi-Fi)	Allows devices to connect without cables
NIC	Enables a device to connect to a network	Unique MAC address; wired or wireless
Firewall	Monitors and controls network traffic	Protects from unauthorised access
Gateway	Connects two different networks	Translates protocols between networks

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3. Detailed Device Descriptions

Modem

Aspect	Description
Definition	A device that connects a network to the internet by modulating/demodulating signals
What it does	Converts digital signals from a computer into analogue signals for transmission over phone/cable lines, and vice versa
Key Feature	Connects to the ISP (Internet Service Provider)
Example	Cable modem, DSL modem
TCP/IP Layer	Network Access Layer (physical connection)

Analogy: A modem is like a **translator**—it converts between the language of the computer (digital) and the language of the phone/cable line (analogue).

Router

Aspect	Description
Definition	A device that directs data between networks and routes traffic
What it does	Determines the best path for data to travel; connects a LAN to the internet
Key Feature	Uses IP addresses to route data
Example	Home Wi-Fi router, enterprise router
TCP/IP Layer	Network Layer (Layer 3) — routing

Analogy: A router is like a **postal sorting office**—it reads the address and decides where to send each package.

Switch

Aspect	Description
Definition	A device that connects devices within a LAN and directs data using MAC addresses
What it does	Forwards data to the correct device within the same network
Key Feature	Uses MAC addresses; more efficient than a hub (only sends to the intended recipient)
Example	Network switch in an office
TCP/IP Layer	Network Access Layer (Data Link Layer) — switching

Analogy: A switch is like a **mail room**—it delivers mail to the right office within the same building without sending it outside.

WAP (Wireless Access Point)

Aspect	Description
Definition	A device that provides wireless connectivity (Wi-Fi) to devices
What it does	Allows devices to connect to a network wirelessly
Key Feature	Extends network coverage; connects wireless devices to the wired network
Example	Wi-Fi access point in a school or office
TCP/IP Layer	Network Access Layer (Physical Layer) — wireless transmission

Analogy: A WAP is like a **radio transmitter**—it broadcasts a signal that wireless devices can pick up.

NIC (Network Interface Card)

Aspect	Description
Definition	Hardware that allows a device to connect to a network
What it does	Provides a physical connection to the network (wired or wireless)
Key Feature	Each NIC has a unique MAC address
Example	Ethernet card, Wi-Fi adapter
TCP/IP Layer	Network Access Layer (Physical/Data Link) — connection

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Firewall

Aspect	Description
Definition	A security device that monitors and controls incoming/outgoing network traffic
What it does	Blocks unauthorised access; allows legitimate traffic
Key Feature	Based on a set of security rules
Example	Hardware firewall, software firewall (Windows Defender)
TCP/IP Layer	Operates at multiple layers (depends on the firewall)

Gateway

Aspect	Description
Definition	A device that connects two different networks
What it does	Translates protocols and formats between networks

Aspect	Description
Key Feature	Often combined with a router in home networks
Example	A device connecting a LAN to a WAN
TCP/IP Layer	Can operate at multiple layers (Application, Network)

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4. TCP/IP Model and Device Mapping

TCP/IP Model Overview

Layer	Function	Devices
Application Layer	Provides services for applications (HTTP, FTP, SMTP)	Firewall (application-level), Gateway
Transport Layer	Ensures reliable data transfer (TCP, UDP)	Firewall
Network Layer	Routes data between networks (IP addressing)	**Router**
Network Access Layer	Physical transmission of data	**Modem, Switch, WAP, NIC**

Data Flow Through Devices

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┌─────────────────────────────────────────────────────────────┐
│                  DATA PACKET JOURNEY                        │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│  1. Device sends data (NIC)                                 │
│               ↓                                             │
│  2. Data goes to Switch (within LAN)                        │
│               ↓                                             │
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│  3. Switch sends to Router (if destined outside LAN)          │
│              ↓                                                │
│  4. Router determines best path (using IP)                    │
│              ↓                                                │
│  5. Data sent to Modem → Internet                             │
│                                                               │
│  Wireless: Device → WAP → Switch → Router → Modem             │
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Common Network Layouts

Home Network:

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    Internet
        │
    [Modem]
        │
    [Router]
        │
    ┌───┴───┐
    │       │
 [Switch] [WAP]
    │       │
    │      [Wireless Devices]
 [Wired Devices]
```

Office Network:

```text

    Internet
        │
    [Firewall]
        │
    [Router]
        │
    [Switch] ── [WAP]
```

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     |  |  |
   [PC] [PC] [Printer]
```

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5. Lesson Sequence

Lesson Plan

Phase	Time	Activity
Engage	5 min	"Network Device Brainstorm"—display home network picture; ask what devices students recognise
Explore	20 min	Core devices (modems, routers, switches, WAPs); "Network Device Charades"
Explain	20 min	TCP/IP model introduction; "Data Packet Journey" animation; Device mapping
Elaborate	10 min	"Network Detective" troubleshooting scenarios
Wrap-up	5 min	Key takeaways; exit ticket

Activity: Network Device Charades

Instructions:

1. Divide students into small groups
2. Each group picks a network device
3. Groups act out the function of the device **without speaking**
4. Other groups guess which device is being acted out

Example Actions:

- **Modem:** Pretend to connect a cable and "translate" between two devices

- **Router:** Pretend to read addresses and send packages in different directions
- **Switch:** Pretend to receive data and deliver it to the correct device
- **WAP:** Pretend to broadcast a signal (arms waving outward)

Activity: Network Detective (Troubleshooting Scenarios)

Instructions: Present a network scenario with a connectivity problem. Students work in pairs/small groups to identify possible causes.

Scenario 1: No Internet Access

- **Problem:** A family is at home, but none of their devices can connect to the internet
- **Possible Causes:** Modem issue, router malfunction, Wi-Fi interference, incorrect Wi-Fi password
- **Guiding Questions:** Is the modem's power light on? Have you tried restarting the modem and router? Are other devices experiencing the same issue?

Scenario 2: Slow Internet Speed

- **Problem:** A student is trying to stream a movie, but the video keeps buffering
- **Possible Causes:** Too many devices using the internet simultaneously; distance from router; ISP congestion
- **Guiding Questions:** How many devices are connected? What is the download speed on a speed test?

Scenario 3: Intermittent Connection

- **Problem:** Internet connection drops out frequently
- **Possible Causes:** Weak Wi-Fi signal; faulty cable; outdated network drivers
- **Guiding Questions:** Does the problem occur in a specific location? Have you tried moving closer to the router?

Scenario 4: Can't Print Wirelessly

- **Problem:** User can connect to Wi-Fi but not print to wireless printer
- **Possible Causes:** Printer not on same network; printer offline; firewall blocking; incorrect drivers
- **Guiding Questions:** Is the printer turned on? Can you ping the printer's IP address?

Scenario 5: Limited Access to Websites

- **Problem:** Employees can access some websites but not others
- **Possible Causes:** Firewall restrictions; content filtering; DNS server issues; ISP blocking
- **Guiding Questions:** Are all employees experiencing the same issue? Can you access blocked websites from outside the office?

6. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions during discussions
Network Device Charades	Assess understanding of device functions through performances
Network Detective	Assess ability to apply knowledge to troubleshooting scenarios
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. What is the main function of a router?
2. What is the difference between a switch and a router?
3. Which TCP/IP layer does a router operate at?
4. Name one device that provides wireless connectivity.

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7. Differentiation

Level	Strategy
Support	Provide visual aids, concise definitions, and real-world examples
Challenge	Encourage advanced students to explore complex scenarios or research emerging network technologies

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8. Resources

- **Presentation** (Slide Deck)

- **eBook** (A2.1.3 with quiz)
- **Worksheet:** Device function matching + diagram drawing
- **Quizlet** (A2.1.3 vocabulary flashcards)

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9. Summary: Key Takeaways

Device	Function
Modem	Connects to the internet (translates signals)
Router	Directs data between networks (uses IP addresses)
Switch	Connects devices within a LAN (uses MAC addresses)
WAP	Provides Wi-Fi connectivity
NIC	Enables a device to connect to a network
Firewall	Monitors and controls network traffic (security)
Gateway	Connects two different networks

TCP/IP Layer Mapping:

- **Router → Network Layer**
- **Switch → Network Access Layer (Data Link)**
- **Modem, WAP, NIC → Network Access Layer (Physical)**

> "*Network devices work together like a postal system—the modem is the connection to the outside world, the router is the sorting office, and the switch is the internal delivery system.*"